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N. NITHISHKUMAR , K.KANISHK , S.SARAYU

Corresponding Author Email ID: vtu28017@veltech.edu.in

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY, AVADI

Design And Development Of Smart Fire Extinguisher Using LoRa

Abstract

Fire accidents in industrial, commercial, and residential environments continue to cause serious loss of life and property, mainly due to delayed fire detection and the absence of immediate response mechanisms. Traditional fire extinguishers require manual operation and do not provide any means to communicate fire incidents to concerned authorities, which limits timely intervention. To overcome these challenges, this project proposes a smart fire extinguisher system enabled with **long-range wireless communication**. The developed system integrates flame, smoke, and temperature sensors with a **microcontroller** to continuously monitor surrounding environmental conditions. When a fire is detected beyond predefined safety limits, the system automatically activates the fire extinguishing mechanism without human intervention. Simultaneously, an alert message containing real-time fire information is transmitted to a remote monitoring station using **LoRa communication technology**. LoRa is selected for its long communication range, low power consumption, and reliable operation in areas with poor or unavailable network connectivity, making it suitable for large industrial premises and remote locations. In addition to automatic fire suppression, the system provides real-time status updates such as fire intensity, extinguisher activation status, and overall system health, enabling safety personnel to respond quickly and effectively. The proposed solution is compact, cost-effective, and easy to deploy, making it suitable for applications in smart buildings, warehouses, and small-scale industries. By combining early fire detection, automatic response, and long-distance communication, the proposed smart fire extinguisher significantly improves fire safety and helps minimize fire-related risks through timely and intelligent intervention.

Key words - LoRa, fire extinguisher